

REAL RAILS // VISUALIZATION AUDIT REPORT (VAR)

PoC #07 · Fraud Velocity Monitor	Archetype: Temporal · Rail: Payment · v2-temporal	Date: 11 Apr 2026 · Sources: CFPB · FRED
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PAS S 18	IMP ROV E 0	FAIL 0	FINAL VERDICT FULL GREEN — 18/18 PASS All fixes applied & verified. Dashboard is client-ready.
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SECTION 1 — Requirement Match (Excel Row vs Final Code)

Criterion	Result	Evidence found in final Repomix / code
Archetype = Temporal (Pattern View)	PASS	Header badge 'ARCHETYPE: TEMPORAL' confirmed in page.tsx. ComposedChart signal stream is the 70% main stage.
Click triangle spike → inspector opens	PASS	handleChartClick(): guards non-anomaly clicks (if !raw?.anomaly return), finds enriched stream point, calls setSelectedAnomaly + scrollIntoView.
Velocity rules feature	PASS	Velocity bar chart with ReferenceLine y=15. 3-color: red≥15/hr, amber≥8, cyan<8. Rule label visible.
Anomaly alerts — 3 spike types	PASS	burst/risk_surge/amount_spike generated in main.py, rendered as colored triangles, documented in sidebar legend, explained in inspector.
IP/device fingerprint placeholders	PASS	device_fingerprint + ip_hash in QueueEvent interface, genSignalStream fallback, inspector 6-card grid, Review Queue table (9 cols). Stream enrichment injects FP/IP when API lacks them.
Risk score trend tab	PASS	RISK SCORE tab: avg_risk area chart + ReferenceLine y=0.55 ALERT threshold + hour-of-day fraud heatmap.
Review queue feature	PASS	REVIEW QUEUE tab: 9-column table (EVENT ID, LABEL, RISK, AMOUNT, VEL/HR, DEVICE, CATEGORY, DEVICE FP, IP HASH). genReviewQueue() provides real fallback data.
Why This Matters panel	PASS	Section B: velocity rule explanation, Real Rails Context callout, FRED DRCCCLACBS 2.61% delinquency widget.
Who Controls the Rail panel	PASS	Section C: 3 rows — Card Networks (Visa/MC/Amex), Regulators (CFPB/FinCEN), Fraud Engines (NICE/Featurespace).
Downloadable sample data	PASS	Section E anchor → /api/download/sample-data. Backend streams 100-row signal stream CSV.
Tooltips on chart hover	PASS	SignalTooltip component renders on hover: anomaly pill, payload values, Max Velocity row. .recharts-wrapper { overflow:visible } prevents clipping.

SECTION 2 — DNA Check (Real Rails Visual Identity Rules)

DNA Rule	Result	Evidence
Background #030712 mandatory	PASS	globals.css --background:#030712. page.tsx root div background:#030712. layout.tsx body bg-[#030712]. Triple confirmed.
70/30 split enforced	PASS	gridTemplateColumns:'70% 30%' in page.tsx. CSS .main-stage-70 and .sidebar-30 utility classes also defined in globals.css.
Sidebar Sections A–E present	PASS	A=KPIs+title, B=Why This Matters+FRED indicator, C=Who Controls, D=Resolution controls, E=CSV export. All 5 confirmed.
Cyan #38BDF8 for active states	PASS	globals.css --primary:#38BDF8. Active tab underline, header dot glow, PEAK VELOCITY KPI all use this value.
Borders #1F2937 / 1px	PASS	globals.css --border:#1F2937. Consistent across all section dividers and card borders in page.tsx.
IBM Plex Mono + Space Grotesk fonts	PASS	globals.css: @import Google Fonts BEFORE @import tailwindcss (BUG FIX #8). font-display:swap included. Body uses Space Grotesk, labels use IBM Plex Mono.
Page title = 'Fraud Velocity Monitor Real Rails'	PASS	layout.tsx metadata: title:'Fraud Velocity Monitor Real Rails'. BUG FIX #10 confirmed.
Archetype badge in header	PASS	'ARCHETYPE: TEMPORAL' indigo pill badge visible in header right side.
MapView.tsx deleted (v1 dead code)	PASS	Directory structure in Repomix: no frontend/app/components/MapView.tsx present. Confirmed deleted.
No hardcoded credentials	PASS	Only process.env.NEXT_PUBLIC_API_URL used. No API keys in any file.
Mock fallback activates automatically	PASS	fetchFallback() wraps all 6 API calls. genSignalStream/genCategoryBreakdown/genReviewQueue/genHeatmap provide real data when backend offline.
Font size readable (not compressed)	PASS	globals.css html{font-size:16px} body{font-size:14px, letter-spacing:normal}. layout.tsx body has text-sm class. BUG FIX #8+#9 confirmed.

SECTION 3 — Data Mapping (CFPB · FRED · Synthetic Stream)

Data Point	Result	Mapping Accuracy
CFPB attribution	PASS	Header: 'CFPB/FRED'. Section B text mentions CFPB complaint data. Section C: CFPB in Regulators row.
FRED economic indicator	PASS	Section B: indigo callout 'FRED · DRCCCLACBS — DELINQUENCY RATE' showing 2.61% with 'Q4 2024 · Federal Reserve FRED (mock)' source label.
Fraud velocity (txns/hr) in 70% stage	PASS	max_velocity drives velocity bar chart with 15/hr rule. Peak Velocity KPI in metric strip. Color coding red/amber/cyan.

Risk score 0–1 in 70% stage	PASS	avg_risk drives RISK SCORE tab area chart. Alert threshold ReferenceLine at y=0.55.
Synthetic labels fraud/suspicious/normal	PASS	Stacked area chart uses all 3 labels with distinct colors. Legend top-right of chart area.
3 anomaly spike types in 70% stage	PASS	burst/risk_surge/amount_spike color-coded on chart markers, documented in sidebar legend, explained with descriptions in inspector.

SECTION 4 — Audit Iteration Log (All Issues Resolved Across Versions)

Fix ID	Severity	Issue	Status	Resolution
Fix 1	FAIL	Page title was 'Create Next App' (default boilerplate)	FIXED	layout.tsx: title → 'Fraud Velocity Monitor Real Rails'
Fix 2	IMPROVE	device_fingerprint + ip_hash not shown in inspector or queue	FIXED	Added to QueueEvent interface, inspector 6-card grid, queue table, stream enrichment
Fix 3	IMPROVE	Google Fonts not loaded — letter-spacing compressed text	FIXED	@import before tailwindcss, font-size:14px, letter-spacing:normal in globals.css
Fix 4	IMPROVE	No FRED indicator in sidebar	FIXED	FRED DRCCCLACBS 2.61% indigo widget added to Section B
Fix 5	IMPROVE	MapView.tsx leftover from v1 Geographic archetype	FIXED	File deleted — confirmed absent from directory structure
Fix 6	BUG	Charts showed width(-1) height(-1) — nothing rendered	FIXED	minHeight:0 on grid, main, and tab content div; flexShrink:0 on header/tabs
Fix 7	BUG	Inspector panel appeared off-screen after click	FIXED	Tab content div: minHeight:'100%' → minHeight:0. Panel now visible in viewport
Fix 8	BUG	handleChartClick opened inspector on every click, not just spikes	FIXED	Added if (!raw?.anomaly) return guard. stream.find() for enriched FP/IP data
Fix 9	BUG	Tooltip clipping at chart edges (overflow:hidden on parent)	FIXED	.recharts-wrapper { overflow: visible !important } added to globals.css
Fix 10	BUG	Review Queue and Category tabs empty — fallback was []	FIXED	genCategoryBreakdown(), genReviewQueue(), genHeatmap() provide real fallback data
Fix 11	BUG	rng() produced negative numbers causing incorrect mock data	FIXED	rng(): s = (Math.imul(1664525,s) + 1013904223) >>> 0 — always unsigned 32-bit

Fix 12	BUG	overflow-x-hidden clipped 9-column Review Queue table	FIXED	layout.tsx body: overflow-x-auto so wide tables scroll instead of clip
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FINAL VERDICT: FULL GREEN — 18 / 18 PASS · 0 IMPROVE · 0 FAIL · CLIENT-READY